

Kaan Aytekin

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EDUCATION

University of Rochester

Rochester, NY

Bachelor of Science, Mechanical Engineering

May 2026

- GPA 3.92 out of 4.00, Dean's List all applicable semesters
- Davis UWC Scholar, Tau Beta Pi Scholar, Helmut Weymann Prize for Engineering Experimentation

ENGINEERING EXPERIENCE (See Portfolio for more detail - www.kaanaytekin.com)

Laboratory for Laser Energetics (DoE funding \$90M+, 400+ employees)

Rochester, NY

Research Assistant

March 2023 - Present

- Conducted in-depth numerical calculations of free-free laser absorption rates in plasmas, incorporating quantum scattering theory and screening effects. Hands on experience in optical experiment setup.
- Leveraged Python programming skills (NumPy, SciPy and Matplotlib), to perform numerical evaluations of Ordinary Differential Equations (ODEs) and integrals.
- Performed computations of 100 absorption correction tables (Gaunt Factor) using a Linux-based HPC cluster. Fitted a parametric analytical expression with a three-variable equation to extract data for integration into opacity and hydro-codes along with fusion experiments.

ASML – Air Bearing Metrology and Testing Rig

Rochester, NY

Team Member

January 2026- May 2026

- Engineered a high-precision metrology rig to characterize load capacity and sub-micron fly-height of commercial air bearings, enabling ASML to evaluate cost-effective off-the-shelf alternatives to custom components.
- Conducted FEA analysis (NASTRAN Sol 101, 103) on load application structure to ensure safety during peak loading conditions.

Newtonian Telescope Optomechanical Design and Manufacture

Rochester, NY

Optomechanical Advisor to Optics Capstone Project

January 2026- May 2026

- Owned the mechanical architecture, precision design, and CNC manufacturing of a Newtonian telescope, engineering mirror mounts and optical alignment mechanisms capable of 0.015-degree tilt increments.
- Performed FEA on primary mirror to model the deflection due to gravity at various telescope angles, fitting surface deformation data with Zernike polynomials to predict and mitigate line-of-sight and alignment errors.
- Conducted structural modal analysis to find the assembly's natural frequencies and mode shapes, optimizing the design to guarantee a fundamental frequency above 120 Hz and effectively eliminate environmental jitter.

UR Baja SAE (Offroad Car Collegiate Design Series)

Rochester, New York

Chief Engineer and Drivetrain Project Team Co-Leader

May 2023 – May 2025

- Directed a 30-member multi-disciplinary team through complete vehicle design and manufacture, achieving a 40% increase in national competition rankings while reducing the project spending by 5%.
- Designed the team's first-ever custom differential along with a custom transfer case and driveshaft using NX.
- Validated gear-train integrity by applying AGMA standards and MATLAB to create solvers for bevel gear forces.
- Optimized axle and universal joint geometry using engineering analysis methods such as FEA (NASTRAN), dynamics simulations (Siemens Simcenter 3D), solid mechanics principles and Design for Manufacturing (DFM), troubleshooting critical interference and loosening issues while achieving a 64% reduction in material costs.
- Manufactured 50+ custom designed axle yokes in a 4-axis CNC mill, used a manual mill to manufacture axle flanges, inserts and more. Designed custom fixture and tooling for sheet metal bending.

Torsion Testing Rig for 4130 Steel Axles

Rochester, NY

Author

January 2024 - May 2024

- Developed a custom torque testing rig and testing methodology to measure the torsional yield characteristics of heat-treated 4130 steel (no-treatment, annealed and quenched samples) using strain gauges, NI data acquisition devices, analyzing data using MATLAB and first principles engineering equations.
- Conclusively validated that targeted heat treatment statistically and significantly improves the torsional yield point of steel half-shaft connections, supporting component weight reduction in vehicle design.

AWARDS

APS DPP '24 Outstanding Poster Award

Atlanta, GA

American Physical Society Division of Plasma Physics Annual Meeting

October 2024

- Awarded to top 5 student presenters out of approx. 150 presenters.

SKILLS & INTERESTS

- **CAD:** Siemens NX, Solidworks, Fusion 360, PTC Creo. **Simulation/CAE:** Simcenter 3D, NASTRAN (linear, nonlinear, real eigenvalues), Multibody Dynamics, ANSYS Fluent (Workbench, Dynamic Meshing, VoF Mixtures)
- **Data Analysis/Coding:** MATLAB (Real Time Data Acquisition, NI, Simulink, Strain Gages), Python (NumPy, SciPy, Matplotlib, Large dataset analysis) **Other:** Opto-Mechanical Design (Portfolio), Test Engineering, Data Acquisition
- Turkish (Native), English (Fluent), German (B1)
- **Music:** played and recorded piano and guitar, led band as the club leader and the lead guitarist.